

Capstone Project Proposal

Name: [REDACTED]

Objective

[Describe the overall objective of the project. If necessary, explain the context and motivation.]

The overall objective of my project is to create a playlist generator that gives exposure to artists and songs not backed by major labels / with less exposure. The motivation behind this is to provide an outlet for music listeners to expand their tastes and find new music as well as provide exposure to smaller artists.

This would be done by having users take a sort of “quiz” in which they choose their favorite artists and genres/subgenres. This will start with the main core labels of electronic, rock, hip-hop, r&b, pop, country, and classical. There will be multiple levels of songs for users to choose from (each category paired with a “I don’t know any of these” button so users can choose to stop their journey based on their music knowledge and preferences). Once the quiz is completed, a 10-15 song long playlist will be generated for the user.

Outcome

[Clearly state the final outcome of the capstone, e.g., predictive model, statistical analysis, dashboard, UI, etc]

The final outcome will be a functional playlist generator with a quiz-style UI. The system will guide users through a structured selection process of genres, subgenres, and favorite artists, then generate a personalized playlist of 10–15 songs by independent or lesser-known artists.

Impact

[In a few sentences, delineate the potential impact of the project. Who would benefit from your deliverables?]

This project addresses the “cold start” problem faced by streaming services by creating an accessible entry point for new users. This tool also aims to maximize user experience by providing recommendations without extensive listening history. Likewise, this actively promotes independent and smaller artists by helping increase streams and visibility within a crowded marketplace. In this way, the project benefits both listeners seeking fresh discoveries and artists striving to reach new audiences.

Dataset(s)

[List your data sources with links to them. If you have already uploaded them to your capstone repository on GitHub, please mention the location. In addition, briefly discuss the datatypes and the reliability of the data.]

[Data from Spotify's Web API](#)

- Spotify audio features

[Data from MusicMap](#) (for genre history)

[Million song data from Last.fm](#) (genre tags and similar songs)

Approach

[Talk about how you plan on approaching this capstone through several steps. List the steps below.]

1. Background research
2. Data collection
3. Data cleaning & exploration
4. Feature engineering /
5. Model development
 - a. Similarity-based methods
 - b. Clustering algorithms
6. Design quiz-style interface
7. Develop playlist generation pipeline

Timeline

[Edit the following example timeline:]

This is a rough timeline for this project:

- (2 Weeks) Background research and literature review on cold start problem and recommendation engines
- (2 Weeks) Data collection and preprocessing
- (3 Weeks) Exploratory data analysis & feature engineering
- (3 Weeks) Model and pipeline development
- (2 Weeks) Quiz interface simulation & playlist integration
- (1 Week) Writing up the report
- (1 Week) Poster and Final Presentation

Possible Issues

[List some of the prospective challenges and issues and discuss how you envision overcome them]

Several challenges may arise throughout this project. For example, data access could be limited by restrictions in Spotify's API, and there may be a presence of inconsistencies within the metadata. This will be addressed by supplementing with other datasets and focusing on representative samples. Likewise, time and scope constraints may prevent development of a polished interface, in which case a simplified quiz simulation will be implemented to ensure the core pipeline is functional.